FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014: 2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE(Advanced), 2015

ANSWERS, HINTS & SOLUTIONS

HALF COURSE TEST –VI
(Paper-2)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	B	B	C
2.	A	D	B
3.	D	B	A
4.	B	C	B
5.	D	C	A
6.	C	B	B
7.	A	A	B
8.	B	C	B
9.	B	C	A
10.	D	C	D
11.	A	B	D
12.	D	C	D
13.	B	C	A
14.	C	C	A
15.	B	A	C
16.	A	C	B
17.	C	B	A
18.	A	A	B
19.	C	C	C
1.	(A) → (q) (B) → (p) (C) → (s) (D) → (r)	(A) → (s) (B) → (r) (C)→ (q) (D) → (p)	(A) → (p, s) (B) → (q, r) (C) → (r) (D) → (p, s)
2.	(A) → (s) (B) → (p) (C) → (q) (D) → (q), (r)	(A) → (s) (B) → (r) (C)→ (p) (D) → (q)	(A) → (q, r, s) (B) → (q) (C) → (r, s) (D) → (p)
3.	(A) → (r) (B) → (p), (q), (r) (C) → (s) (D) → (r)	(A) → (r) (B) → (p) (C) → (p, s) (D) → (p, q)	(A) → (s) (B) → (r) (C) → (p) (D) → (q)

Physics

PART – I

SECTION – A

5. Use superposition principle

6. $1000 = \frac{v}{4\ell_1} = \frac{3v}{4\ell_2}$

$\Rightarrow \quad \ell_2 = 3\ell_1$

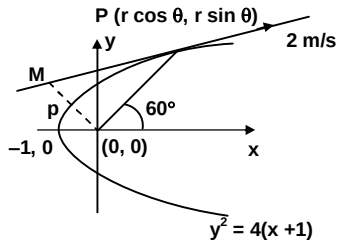
$\Rightarrow \quad \ell_1 = 8$ cm and $\ell_2 = 24$ cm

$\Rightarrow$ minimum volume $= 16 \times 100 = 1600\ cm^3$

14. $r^2 \frac{d\theta}{dt} = vp$

15. $\omega = \frac{\bar{v}_A - \bar{v}_B}{r} = \frac{10-15}{5} = -1$ rad/sec.

16. $r \cos\theta + 2 = r$



$\frac{2}{r} = 1 - \cos\theta = 2\sin^2\frac{\theta}{2}$

$\frac{1}{r} = \sin^2\frac{\theta}{2}$

$p = \sqrt{ar}$ [property of parabola, here a = 1]

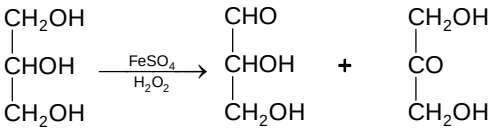
$\omega = \frac{d\theta}{dt} = \frac{vp}{r^2} = \frac{v\sqrt{r}}{r^2} = \frac{v}{r^{3/2}} = 2 \times \left(\frac{1}{2}\right)^3$

Chemistry

PART – II

SECTION – A

1. Salts of soft acid with soft base are more stable than soft acid and hard base.

2. 

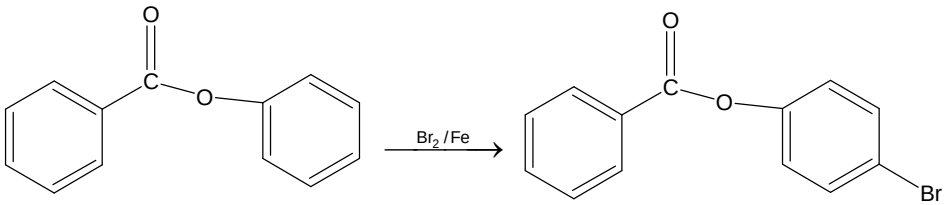
Glyceraldehyde Dihydroxyacetone

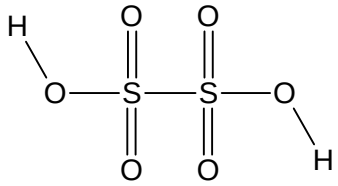
3. Half-life is independent of concentration of Zn, thus order w.r.t. [Zn] = 1
At pH = 2, $[H^+] = 10^{-2}$, $t_{1/2}$ = 10 minutes
pH = 3, $[H^+] = 10^{-3}$, $t_{1/2}$ = 100 minutes

$$\frac{[t_{1/2}]_1}{[t_{1/2}]_2} = \left(\frac{a_2}{a_1}\right)^{n-1}$$

$$\frac{10}{100} = \left(\frac{10^{-3}}{10^{-2}}\right)^{n-1}$$

$\therefore$ n = 2, thus w.r.t. $[H^+]$ order = 2
Rate law = $k\,[H^+]^2\,[Zn]$
Hence, (B).

4. 

5. 

6. $$Y_B = \frac{P_B^o X_B}{P_{Total}}$$

Graph Y_B vs $\frac{1}{P_{Total}}$ is linear.

7. $$P_c V_c = \frac{3}{8} R T_c$$

i.e. $\frac{8}{3}$ times of $\frac{RT}{PV}$ at normal condition

8. $O_3 \rightleftharpoons O_2 + O$ (fast)

$$K_{eq.} = \frac{[O_2][O]}{[O_3]} \quad \ldots(1)$$

From Eq. (1), $[O] = \frac{K_{eq.}[O_3]}{[O_2]}$

$O + O_3 \xrightarrow{slow} 2O_2$

$\therefore Rate_1 = k[O][O_3]$

$$Rate_1 = k.K_{eq.}\frac{[O_3]}{[O_2]} \times [O_3]$$

$$Rate_1 = \frac{k'[O_3]^2}{[O_2]}$$

$$Rate_2 = \frac{k'[O_3]^2}{\frac{[O_2]}{2}} = 2k'$$

Mathematics

PART – III

SECTION – A

1. Sum of coefficients in $(1 - x \sin\theta + x^2)^n$ is $(1 - \sin\theta + 1)^n$ (putting x = 1)
This sum is greatest when $\sin\theta = -1$, then maximum sum is 3^n

2. Let x = p – 5, y -= q – 5 and z = r – 5, where p, q, r ≥ 0
Then the given equation reduces to p + q + r = 15
The total number of solutions to this is ${}^{17}C_2 = 136$

3. Let $A \equiv (at^2, 2at)$, $B \equiv (at^2, -2at)$

$m_{OA} = \frac{2}{t}$, $m_{OB} = -\frac{2}{t}$

Thus, $\left(\frac{2}{t}\right)\left(-\frac{2}{t}\right) = -1 \Rightarrow t^2 = 4$

Thus, tangents will intersect at (–4a, 0)

4. Let directrix be $x = \frac{a}{e}$ and focus be S(ae, 0). Let P(a sec θ, b tan θ) be any point on the curve

Equation of tangent at P is $\frac{x\sec\theta}{a} - \frac{y\tan\theta}{b} = 1$

Let F be the intersection point of tangent at directrix

Then, $F \equiv \left(\frac{a}{e}, \frac{b(\sec\theta - e)}{e\tan\theta}\right)$

$\Rightarrow m_{SF} = \frac{b(\sec\theta - e)}{-e\tan\theta(a^2 - 1)}$, $m_{PS} = \frac{b\tan\theta}{a(\sec\theta - e)}$

$\Rightarrow m_{SF} \cdot m_{PS} = -1$

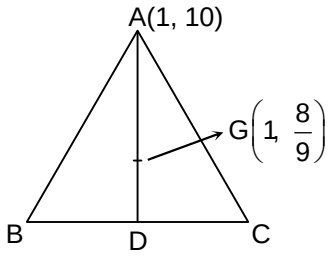
5. Circumcentre $O \equiv \left(-\frac{1}{3}, \frac{2}{3}\right)$ and

Orthocentre $H \equiv \left(\frac{11}{3}, \frac{4}{3}\right)$

Hence, co-ordinates of G are $\left(1, \frac{8}{9}\right)$

Now, AD : DG = 3 : 1

Hence, the co-ordinates of $D \equiv \left(1, -\frac{11}{3}\right)$



6. $\cos^3 x \sin 2x = \cos^2 x \cdot \cos x \cdot \sin 2x$

$= \left(\frac{1-\cos 2x}{2}\right)\left(\frac{2\sin 2x\cos x}{2}\right)$

$= \frac{1}{4}(1-\cos 2x)(\sin 3x + \sin x)$

$= \frac{1}{4}\left[\sin 3x + \sin x - \frac{1}{2}(2\sin 3x \cdot \cos 2x) - \frac{1}{2}(2\cos 2x \sin x)\right]$

$= \frac{1}{4}\left[\sin x + \frac{1}{2}\sin 3x - \frac{1}{2}\sin 5x\right]$

$\Rightarrow a_1 = \frac{1}{4}$, $a_3 = \frac{1}{8}$, n = 5

7. According to the question, 1 must lie between the roots
Since, $4p - p^2 - 5 < 0$, thus, $f(1) > 0$
$\Rightarrow p^2 - 5p + 4 < 0$
$\Rightarrow 1 < p < 4$
$\Rightarrow p \in \{2, 3\}$

8. Let $z = x + iy$. Then, $x = \lambda + 3$ and $y = -\sqrt{5 - \lambda^2}$
$\Rightarrow (x - 3)^2 = \lambda^2$ and $y^2 = 5 - \lambda^2$
Thus, $(x - 3)^2 = 5 - y^2 \Rightarrow (x - 3)^2 + y^2 = 5$
It is a semicircle as $y < 0$

9. We have $\frac{\pi}{4} = \left(1 - \frac{1}{3}\right) + \left(\frac{1}{5} - \frac{1}{7}\right) + \left(\frac{1}{9} - \frac{1}{11}\right) + ...$
$= \frac{2}{1\times 3} + \frac{2}{5\times 7} + \frac{2}{9\times 11} + ...$
$\Rightarrow \frac{1}{1\times 3} + \frac{1}{5\times 7} + \frac{1}{9\times 11} + ... = \frac{\pi}{8}$

10. $30 - 9\pi \in [0, \pi]$ is true but it is not principal value of $\cos^{-1}(\cos 30)$
as $\cos^{-1}(\cos 30) = \cos^{-1}(\cos(9\pi + (30 - 9\pi)))$
$= \cos^{-1}(-\cos(30 - 9\pi)) = \pi - (30 - 9\pi) = 10\pi - 30$

11. Locus of point of intersection of perpendicular tangents is director circle
If there exists exactly one such point on the line $3x + 4y + 5\sqrt{5} = 0$, then it must touch the director circle $x^2 + y^2 = a^2 + 1$
$\Rightarrow 5 = a^2 + 1$
$\Rightarrow a^2 = 4$
$\Rightarrow a = 2$
Hence, eccentricity = $\sqrt{1 - \frac{1}{4}} = \frac{\sqrt{3}}{2}$

12. According to given data, $2a - 5 + a^2 - 3 < 0$
$\Rightarrow a^2 + 2a - 8 < 0$
$\Rightarrow a \in (-4, 2)$

13. Given equation is $px^2 + qx + r = 0$
Let, $f(x) = px^2 + qx + r$
$f(0) = r > 0$
$f(1) = p + q + r < 0$
$f(-1) = p - q + r < 0$
Hence, one root lies in $(-1, 0)$ and the other in $(0, 1)$
$\therefore [\alpha] = -1$ and $[\beta] = 0$
$\Rightarrow [\alpha] + [\beta] = -1$

14.-16. Let m and (m + 1) be the removed numbers from 1, 2, ..., n
Then, sum of the remaining numbers is $\frac{n(n+1)}{2} - (2m + 1)$
From given conditions, $\frac{105}{4} = \frac{\frac{n(n+1)}{2} - (2m+1)}{n - 2}$
$\Rightarrow 2n^2 - 103n - 8m + 206 = 0$
Since n and m are integers, so n must be even
Let $n = 2k$. Then, $m = \frac{4k^2 + 103(1 - k)}{4}$
Since m is an integer, then $1 - k$ must be divisible by 4. Let $k = 1 + 4t$. Then we get $n = 8t + 2$ and $m = 16t^2 - 95t + 1$

Now, $1 < m < n$
$\Rightarrow 1 \le 16t^2 - 95t + 1 < 8t + 2$
Solving, we get $t = 6$. Hence, $n = 50$ and $m = 7$
Hence, the removed numbers are 7 and 8
Also, sum of all numbers is $\frac{50(50+1)}{2} = 1275$

17.-19. Suppose A contains r $(0 \le r \le n)$ elements
Then, B is constructed by selecting some elements from the remaining n – r elements
Here, A can be chosen in ${}^nC_r$ ways and B in ${}^{n-r}C_0 + {}^{n-r}C_1 + \ldots + {}^{n-r}C_{n-r} = 2^{n-r}$ ways
So, the total number of ways of choosing A and B is ${}^nC_r \times 2^{n-r}$. But r can vary from 0 to n

So, total number of ways is $\sum_{r=0}^{n} {}^nC_r \times 2^{n-r} = (1+2)^n = 3^n$

If A contains r elements, then B contains (r + 1) elements
Then, the number of ways of choosing A and B is ${}^nC_r \times {}^nC_{r+1}$
But r can vary from 0 to (n – 1)

So, the number of ways is $\sum_{r=0}^{n-1} {}^nC_r\,{}^nC_{r+1} = {}^nC_0\,{}^nC_1 + {}^nC_1\,{}^nC_2 + \ldots + {}^nC_{n-1}\,{}^nC_n = {}^{2n}C_{n-1}$

Let A contains $r(0 \le r \le n)$ elements
Then, A can be chosen in ${}^nC_r$ ways. The subset B of A can have at most r elements, and the number of ways of choosing B is 2^r. Therefore, the number of ways of choosing A and B is ${}^nC_r \times 2^r$. But r can vary from 0 to n

So, the total number of ways is $\sum_{r=0}^{n} {}^nC_r \times 2^r = (1+2)^n = 3^n$

SECTION – B

1. (A) We must have, $e_1 < 1 < e_2$
$\Rightarrow f(1) < 0$
$\Rightarrow 1 - a + 2 < 0$
$\Rightarrow a > 3$

(B) We must have both roots greater than 1
(i) $D > 0$ or $a^2 - 4 > 0$ or $a \in (-\infty, -2) \cup (2, \infty)$
(ii) $1 \cdot f(1) > 0$ or $1 - a + 2 > 0$ or $a < 3$
(iii) $\frac{a}{2} > 1 \Rightarrow a > 2$
Thus, we have $a \in (2, 3)$

(C) We must have $\frac{1}{e_1^2} + \frac{1}{e_2^2} = 1$

$\Rightarrow \frac{(e_1 + e_2)^2 - 2e_1e_2}{e_1^2e_2^2} = 1$

$\Rightarrow \frac{a^2 - 4}{4} = 1$

$\Rightarrow a = \pm 2\sqrt{2}$

(D) We must have $e_2 < \sqrt{2} < e_1$

$\Rightarrow f(\sqrt{2}) < 0$

$\Rightarrow 2 - a\sqrt{2} + 2 < 0$

$\Rightarrow a > 2\sqrt{2}$

2. (A) $\left(\sin^{-1}x\right)^2+\left(\sin^{-1}y\right)^2=\frac{\pi^2}{2}$

$\Rightarrow \left(\sin^{-1}x\right)^2=\left(\sin^{-1}y\right)^2=\frac{\pi^2}{4}$

$\Rightarrow \sin^{-1}x=\pm\frac{\pi}{2},\ \sin^{-1}y=\pm\frac{\pi}{2}$

$\Rightarrow x = \pm1$ and $y = \pm 1$

$\Rightarrow x^3 + y^3 = -2, 0, 2$

(B) $\left(\cos^{-1}x\right)^2+\left(\cos^{-1}y\right)^2=2\pi^2$

$\Rightarrow \left(\cos^{-1}x\right)^2=\left(\cos^{-1}y\right)^2=\pi$

$\Rightarrow x = y = -1$

$\Rightarrow x^5 + y^5 = -2$

(C) $\left(\sin^{-1}x\right)^2\left(\cos^{-1}y\right)^2=\frac{\pi^4}{4}$

$\Rightarrow \left(\sin^{-1}x\right)^2=\frac{\pi^2}{4}$ and $\left(\cos^{-1}y\right)^2=\pi^2$

$\Rightarrow \left(\sin^{-1}x\right)=\pm\frac{\pi}{2}$ and $\left(\cos^{-1}y\right)=\pi$

$\Rightarrow x = \pm1$ and $y = -1$

$\Rightarrow |x - y| = 0, 2$

(D) $|\sin^{-1} x - \sin^{-1} y| = \pi$

$\Rightarrow \sin^{-1}x=-\frac{\pi}{2}$ and $\sin^{-1}y=\frac{\pi}{2}$

or $\sin^{-1}x=\frac{\pi}{2}$ and $\sin^{-1}y=-\frac{\pi}{2}$

$\Rightarrow x^y = 1^{(-1)}$ or $(-1) = 1$ or -1

3. (A) $z^4 - 1 = 0$

$\Rightarrow z^4 = 1 = \cos 0 + i \sin 0$

$\Rightarrow z = (\cos 0 + i \sin 0)^{1/4} = \cos 0 + i \sin 0$

(B) $z^4 + 1 = 0$

$\Rightarrow z^4 = -1 = \cos \pi + i \sin \pi$

$\Rightarrow z=(\cos\pi+i\sin\pi)^{1/4}=\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}$

(C) $iz^4 + 1 = 0$

$\Rightarrow z^4 = i = \cos\frac{\pi}{2}+i\sin\frac{\pi}{2}$

$\Rightarrow z=\left(\cos\frac{\pi}{2}+i\sin\frac{\pi}{2}\right)^{1/4}=\cos\frac{\pi}{8}+i\sin\frac{\pi}{8}$

(D) $iz^4 - 1 = 0$

$z^4 = -i = \cos\frac{\pi}{2}-i\sin\frac{\pi}{2}$

$\Rightarrow z=\left(\cos\frac{\pi}{2}-i\sin\frac{\pi}{2}\right)^{1/4}=\cos\frac{\pi}{8}-i\sin\frac{\pi}{8}$